

```
# Neural Network Analysis for Two-Class Data
```

```
import numpy as np
```

```
from sklearn.datasets import make_classification
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.neural_network import MLPClassifier
```

```
from sklearn.metrics import accuracy_score, classification_report
```

```
# Create Two-Class Dataset
```

```
X, y = make_classification(
```

```
    n_samples=500,
```

```
    n_features=4,
```

```
    n_classes=2,
```

```
    n_clusters_per_class=1,
```

```
    random_state=42
```

```
)
```

```
# Split the dataset into Training and Testing data
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
# Feature Scaling
```

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

```
# Create Neural Network Model
```

```
model = MLPClassifier(  
    hidden_layer_sizes=(2, 2, 2), # 3 Hidden Layers, 2 Neurons each  
    activation='relu',           # ReLU Activation Function  
    learning_rate_init=0.001,    # Learning Rate  
    max_iter=1000,  
    random_state=42  
)
```

```
# Train the Model
```

```
model.fit(X_train, y_train)
```

```
# Make Predictions
```

```
y_pred = model.predict(X_test)
```

```
# Calculate Accuracy
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
# Display Results
```

```
print("Neural Network Analysis for Two-Class Data")
```

```
print("-----")
```

```
print("Learning Rate: 0.001")
```

```
print("Activation Function: ReLU")
```

```
print("Hidden Layers: 3")
```

```
print("Hidden Neurons: 2")
```

```
print("-----")
```

```
print("Accuracy:", accuracy)
```

```
print("\nClassification Report:")  
print(classification_report(y_test, y_pred))
```

Output

```
.. Neural Network Analysis for Two-Class Data  
-----  
Learning Rate: 0.001  
Activation Function: ReLU  
Hidden Layers: 3  
Hidden Neurons: 2  
-----  
Accuracy: 0.9  
  
Classification Report:  
              precision    recall  f1-score   support  
  
     0           0.97       0.80       0.88         46  
     1           0.85       0.98       0.91         54  
  
   accuracy              0.90              100  
  macro avg           0.91       0.89       0.90       100  
weighted avg           0.91       0.90       0.90       100
```